CO3 - Home Assignment 2420030233

1. Verify the KKT conditions for the following constrained optimisation problem.

$\max, f(x_1, x_2) = x_1^2 + x_2^2 - 4x_1 - 6x_2$

s.to. $x_1 + x_2 \leq 6$

$x_1 + 2x_2 \leq 10$

$x_1, x_2 \geq 0$

Sol:- Opt, $z = f(x) = x_1^2 + x_2^2 - 4x_1 - 6x_2$

$g(x) = x_1 + x_2 - 6$, $h(x) = x_1 + 2x_2 - 10$

Type II: 2 variables and 2 constraints

$L(x, \lambda_1, \lambda_2) = f(x) - \lambda_1 g(x) - \lambda_2 h(x)$

$L(x, \lambda_1, \lambda_2) = [x_1^2 + x_2^2 - 4x_1 - 6x_2] - \lambda_1 [x_1 + x_2 - 6]$
$- \lambda_2 [x_1 + 2x_2 - 10]$

$\frac{\partial L}{\partial x_1} = (2x_1 - 4) - \lambda_1 - \lambda_2$ — (1)

$\frac{\partial L}{\partial x_2} = (2x_2 - 6) - \lambda_1 - 2\lambda_2$ — (2)

$\lambda_1 (g) = 0 \Rightarrow \lambda_1 (x_1 + x_2 - 6)$ — (3)

$\lambda_1 (h) = 0 \Rightarrow \lambda_1 (x_1 + 2x_2 - 10)$ — (4)

~~$x_1 + x_2 - 6 \leq 0$ — (5)~~

~~$x_1 + 2x_2 - 10 \leq 0$ — (6)~~

$x_1, x_2 \geq 0$ — (7)

case(i):- when $\lambda_1 = 0, \lambda_2 = 0$

(1) $2x_1 - 4 \Rightarrow 2x_1 = 4$ $\boxed{x_1 = 2}$

(2) $2x_2 - 6 \Rightarrow 2x_2 = 6$ $\boxed{x_2 = 3}$

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Verifications:-

$x_1+x_2-6 \Rightarrow 2+3-6 \Rightarrow 5-6 \Rightarrow -1 \le 0$ (✓)

$x_1+2x_2-10 \Rightarrow 2+6-10 \Rightarrow -2 \le 10$ (✓)

$x_1, x_2 \ge 0$ — (✓)

$2x_1-4-\lambda_1-\lambda_2=0 \Rightarrow \boxed{\lambda_1=-\lambda_2}$

$2x_2-6-\lambda_1-2\lambda_2=0 \Rightarrow -\lambda_1-2\lambda_2=0$

$+\lambda_2-2\lambda_2 \Rightarrow 0 \quad -\lambda_2=0$

$\lambda_2=0$

$\max, Z = 2^2+3^2-4(2)-6(3) \Rightarrow 4+9-8-18$

$\Rightarrow 5-18=-13.$

2. Solve the following NLPP using Beal's method.

$\min, f(x_1,x_2)=2x_1^2+x_2^2+2x_1x_2-10x_1-10x_2$

s.to. $x_1+x_2 \le 10$

$x_1, x_2 \ge 0$

Sol: $\max, z^* = f(x_1,x_2)=2x_1^2+x_2^2+2x_1x_2-10x_1-10x_2$

$\max, z^* = f(x_1,x_2)=-2x_1^2-x_2^2-2x_1x_2+10x_1+10x_2$

s.to.c. $x_1+x_2-s_1=10$

$x_1, x_2 \ge 0$

select m variables as BV,

$m=1 \quad n=3$

$n-m=3-1=2$

BV

(x_1,x_2) express s_1 and $f(x)$ in the form of x_1, x_2

$s_1=x_1+x_2-10$

$\max, z=f(x)=10x_1+10x_2-2x_1x_2-2x_1^2-x_2^2$

$\frac{\partial f}{\partial x_1}=10-2x_2-4x_1 \qquad \frac{\partial f}{\partial x_2}=10-2x_1-2x_2$

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$\left(\frac{\partial f}{\partial x_1}\right)(x_1,x_2) = 10 \quad \left(\frac{\partial f}{\partial x_2}\right)(x_1,x_2) = 10$

most +ve $= 10$.

$x_1 + x_2 \leq 10$

entering $\Rightarrow x_2$

Ratio $= \dfrac{\text{RHS}}{\text{coeff and entering variable}}$

$\min\left\{\frac{10}{|-2|}, \frac{10}{|-2|}\right\}$ $\quad x_1 = 0, x_2 = 0$

$x_1 + x_2 \leq 10 \Rightarrow x_2 \leq 10$

$x_2 = 1$

$(x_1, x_2) = (2,3)$

constraint, $2+3 = 5 \leq 6$ ✓

$2 + 2(3) = 8 \leq 10$ ✓

$f(2,3) = 13 - 26 - 8 \Rightarrow -13$

max, $f(2,3) = -13$.

3> Use wolfe's method, to min, $f(x_1,x_2) = 2x_2^2 + x_1^2 - 2x_1x_2 - 4x_1$ s.to.c $x_1 + x_2 \leq 4$

$x_1, x_2 \geq 0$

Sol:- min, $f(x) = x_1^2 + 2x_2^2 - 2x_1x_2 - 4x_1$

s.to.c.

$x_1 + x_2 \leq 4$, $x_1, x_2 \geq 0$ $\quad -x_1 \leq 0$

$-x_2 \leq 0$

max, $Z^* = -x_1^2 - 2x_2^2 + 2x_1x_2 + 4x_1$

2> max, $Z^* = -x_1^2 - 2x_2^2 + 2x_1x_2 + 4x_1$

s.to.c $x_1 + x_2 + s_1^2 = 4$

$-x_2 + s_2^2 = 0 \quad g_1(x) = x_1 + x_2 + s_1^2 - 4 = 0$

$-x_2 + s_3^2 = 0 \quad g_2(x) = -x_1 + s_2^2 = 0$

$g_3(x) = -x_2 + s_3^2 = 0$

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3> lagrangian.

$L(x, \lambda_1, \lambda_2, \lambda_3) = f(x) - \lambda_1 g_1(x) - \lambda_2 g_2(x) - \lambda_3 g_3(x)$

$= (-x_1^2 - 2x_2^2 + 2x_1x_2 + 4x_1) - \lambda_1 (x_1 + x_2 + s_1^2 - 4)$
$- \lambda_2(-x_1 + s_2^2) - \lambda_3 (-x_2 + s_3^2)$

$\frac{\partial L}{\partial x_1} = -2x_1 + 2x_2 + 4 - \lambda_1 + \lambda_2$

$\frac{\partial L}{\partial x_2} = -4x_2 + 2x_1 - \lambda_1 + \lambda_3$

$\frac{\partial L}{\partial \lambda_1} = -(x_1 + x_2 + s_1^2 - 4)$

$\frac{\partial L}{\partial \lambda_2} = -(-x_1 + s_2^2)$ $\frac{\partial L}{\partial \lambda_3} = -(-x_2 + s_3^2)$

4> modified LPP,

$-2x_1 + 2x_2 + 4 - \lambda_1 + \lambda_2 = 0$

$-2x_1 + 2x_2 - \lambda_1 + \lambda_2 = -4$

$2x_1 - 2x_2 + \lambda_1 - \lambda_2 = 4$ — (1)

$2x_1 - 4x_2 - \lambda_1 + \lambda_3 = 0$ — (2)

$x_1 + x_2 + s_1^2 = 4$ — (3)

6> 2 phase simplex method.

obj, $Z = -A_1 - A_2$

$x_1 + x_2 + s_1^2 = 4$

7>

		c_j	0	0	0	0	0	0	-1	-1
BV	CB	x_B	x_1	x_2	λ_1	λ_2	λ_3	s_1^2	A_1	A_2
A_1	-1	4	2	-2	1	-1	0	0	1	0
A_2	-1	0	2	-4	-1	0	1	0	0	1
s_1^2	0	4	1	1	0	0	0	1	0	0
			-4↑	6	0	1	-1	0	-2	-2

$R_1 = \frac{R_1}{2}$ $\quad R_2 \to R_2 - 2R_1$

		c_j	0	0	0	0	0	0	-1	-1
BV	CB	x_B	x_1	x_2	λ_1	λ_2	λ_3	s_1^2	A_1	A_2
x_1	0	4	1	1	0	0	0	1	0	0
A_2	1	8	0	1	-3/2	0	-1	2	0	-1
λ_2	0	4	0	4	1	1	0	2	-1	0
$\Delta_j = Z_j - C_j$			0	1	-3/2	0	-1	2	-1	0
$\Rightarrow 8$										

$x_1 = 2.8$, $x_2 = 1.2$ $\quad f_{min} = -7.2$.

4) Formulate that dual of the following G.P and find optimal soln.

$\min, f(x_1, x_2) = 6x_1 + x_1^{-2} x_2^{-1} + 2x_2,$

$x_1, x_2 \geq 0$

Sol:- $T = 3$, $N = 2$

$DOD = 0 \geq 0$ unique soln.

let $f(x) = U_1 + U_2 + U_3$

$U_1 = 6x_1$, $U_2 = x_1^{-2} x_2^{-3}$ $U_3 = 2x_2$

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$$f(x) \geq \prod_{i=1}^{n} \left(\frac{U_i}{\delta_i}\right)^{\delta_i}$$

$$\geq \left(\frac{U_1}{\delta_1}\right)^{\delta_1} \left(\frac{U_2}{\delta_2}\right)^{\delta_2} \left(\frac{U_3}{\delta_3}\right)^{\delta_3}$$

$$\geq \left(\frac{6x_1}{\delta_1}\right)^{\delta_1} \left(\frac{x_1^{-2} x_2^{-3}}{\delta_2}\right)^{\delta_2} \left(\frac{2x_2}{\delta_3}\right)^{\delta_3}$$

$$\geq x_1^{\delta_1 - 2\delta_2} \cdot x_2^{-3\delta_2 + \delta_3} \left(\frac{6}{\delta_1}\right)^{\delta_1} \left(\frac{1}{\delta_2}\right)^{\delta_2} \left(\frac{2}{\delta_3}\right)^{\delta_3}$$

$\delta_1 - 2\delta_2 = 0$, $\quad -3\delta_2 + \delta_3 = 0$

$\delta_1 = 2\delta_2$ $\qquad 3\delta_2 = \delta_3$

w.k.t. $\delta_1 + \delta_2 + \delta_3 = 1$ $\qquad 2\delta_2 + \delta_2 + 3\delta_2 = 1$

$\delta_2 = \frac{1}{6}$, $\quad \delta_1 = \frac{1}{3}$, $\quad \delta_3 = \frac{1}{2}$

$$f(x) \geq x_1^0 \cdot x_2^0 \left[\frac{6}{1/3}\right]^{1/3} \left[\frac{1}{1/6}\right]^{1/6} \left[\frac{2}{1/2}\right]^{1/2}$$

$$\geq [18]^{1/3} [6]^{1/6} [4]^{1/2}$$

≈ 7.07 max,

$$\frac{U_1}{\delta_1} = \frac{U_2}{\delta_2} = \frac{U_3}{\delta_3} \Rightarrow \frac{6x_1}{1/3} = \frac{x_1^{-2} x_2^{-3}}{1/6} = \frac{2x_2}{1/2}$$

$$\frac{3}{x_1^2 x_2^4} = 2 \Rightarrow 9x_1 = 2x_2$$

$x_1 = 0.393$, $\quad x_2 = 1.768$, $\quad$ ~~x_3~~

5) Solve the following GP

$$\min, f(x_1, x_2, x_3) = 40 x_1^{-1} x_2^{-1} + 20 x_1 x_3^{-1} + 20 x_2 x_3 \qquad x_1, x_2, x_3 \geq 0$$

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Sol:- $T = 3$, $W = 3$

$DOD = 3 - 3 - 1 = -1$

$DOD < 0$

$\therefore$ Since $DOD < 0$ no solution.